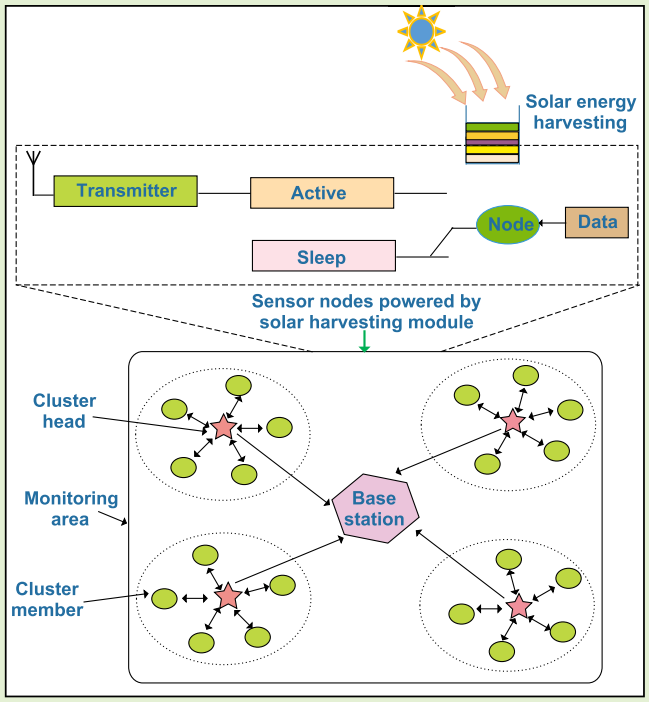
An Efficient Routing Awareness Based Scheduling Approach in Energy Harvesting Wireless Sensor Networks

Dipak Kumar Sah, *Student Member, IEEE*, Abhishek Hazra, *Graduate Student Member, IEEE*, Nabajyoti Mazumdar, *Member, IEEE*, and Tarachand Amgoth, *Member, IEEE*

***Abstract*—Wireless sensor networks (WSNs) have emerged as a promising technology for various Internet of Things (IoT)-based sensing applications. Generally, sensor nodes are powered by nonrechargeable and low-capacity batteries; they only last for a small amount of time. Due to their dispersed nature, sensor nodes cannot have their batteries replaced or maintained locally. The energy harvesting (EH) concept shows commitment as a method for addressing the current energy shortage. Another problem is the intra-energy control of various network load functions, including coverage preservation, routing, clustering, etc. This study proposes an efficient routing awareness scheduling (ERAS) algorithm to address the abovementioned issues. The ERAS method employs a solar harvesting system as a hierarchical clustering-based routing protocol. The presented scheme dynamically alters nodes between states and adjusts to new states based on the insides of sensed data packets. Synchronization-based scheduling ensures proper coordination between the cluster head (CH) and cluster members (CMs). It selects the CH based on both remaining and harvested energy and the number of active nodes to improve throughput. Surprisingly proposed EASR algorithm performs efficiently while considering various performance matrices, including network lifetime, packet delivery ratio (PDR), energy consumption, and network sustainability. Extensive simulation results exhibit the network lifetime of the proposed ERAS technique (79.8%) compared to energy routing awareness (ERA), sleep-wake energy-efficient distributed (SEED) and cross-layer design with energy scavenging and transfer (CREST) methods as 67.3%, 73.8%, and 75.2%, respectively.**



***Index Terms*—Routing awareness, scheduling method, solar energy harvesting (EH), wireless sensor networks (WSNs).**

Manuscript received 20 January 2023; revised 5 May 2023; accepted 11 May 2023. Date of publication 12 June 2023; date of current version 1 August 2023. This work was supported in part by the DST-SERB, India, under Empowerment and Equity Opportunities for Excellence in Science under Grant EEQ/2022/000400. The associate editor coordinating the review of this article and approving it for publication was Prof. Jaime Lloret. *(Corresponding author: Tarachand Amgoth.)*

Dipak Kumar Sah is with the Department of Computer Engineering and Applications, GLA University, Mathura, Uttar Pradesh 281406, India, and also with the IIT (ISM) Dhanbad, Dhanbad, Jharkhand 826004, India (e-mail: dipak.sah@gla.ac.in; dksah.iitism@gmail.com).

Abhishek Hazra is with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore (e-mail: hazra@nus.edu.sg).

Nabajyoti Mazumdar is with the Department of Information Technology, IIIT, Allahabad, Allahabad 211012, India (e-mail: nabajyoti@iiita.ac.in).

Tarachand Amgoth is with the IIT (ISM) Dhanbad, Dhanbad, Jharkhand 826004, India (e-mail: tarachand.ism@gmail.com).

Digital Object Identifier 10.1109/JSEN.2023.3279249

Nomenclature

N	Number of nodes in the sensor network.
r	Number of rounds.
E_{har}	Energy harvesting rate in different rounds [mW].
E_{res}	Residual energy of sensor node [J].
E_{diss}	Energy dissipated for various loads operation [J].
l	Length of data packets.
E_{DA}	Energy is taken for data aggregation [J].
$E_{\text{TDMA}}^{\text{total}}$	Total energy consumption for time division multiple access (TDMA) scheduling [J].
$E_{\text{Intra}}^{\text{routing}}$	Energy dissipated for intra cluster routing [J].
$E_{\text{Inter}}^{\text{routing}}$	Energy consumed for inter cluster routing [J].

1558-1748 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Indian Institute of Information technology Sricity. Downloaded on January 19,2026 at 12:53:14 UTC from IEEE Xplore. Restrictions apply.

E_{CH} CH spends energy sending data to the BS [J].
E_{CM} CM consumes energy to forward information to the CH [J].
E_{elec} energy dissipated of the electronics circuit [J].

I. Introduction

IN THE age of super connectivity, physical objects connect, gather real-time information from various sources, and transfer the data back into the environment [1]. Users can access resources without comprehending the complicated composite structure of heterogeneous systems and hybrid Internet of Things (IoT) infrastructure [2]. In the IoT paradigm, wireless sensor networks (WSNs) are used to collect, manage, and process data. WSNs are comprised of low-cost sensing devices capable of reading, computing, and communicating data [3], [4]. Due to benefits such as low-power signal processing, pervasive surveillance, and cost-effective large-scale deployment, WSNs have been used in various applications, including military surveillance, civilian environmental monitoring, etc. The sensor node devices contain four essential components: a processing unit, a sensing unit, a storage unit, and a transceiver unit. The power unit is critical because each network load function leans on the sensor node battery's energy level. In general, sensor nodes are dispersed randomly, including any inaccessible location, unknown area, hill, forest area, and so on. However, the primary task is to ensure that all sensing information is routed to the sink node via various intermediate nodes [5]. In this context, several studies have been conducted to preserve sensor node energy resources through a routing-based clustering approach. In clustered architecture, the nodes are divided into many clusters, each with one cluster head (CH) and some local members. The cluster architecture has been demonstrated to scale effectively to a self-organized WSN with varying network sizes and sensor density [6]. Nonrechargeable, finite-capacity batteries are frequently used to power sensor nodes in WSNs. Because of the remote deployment of sensor nodes, battery replacement will be time-consuming and costly [7]. As a result, the long-term viability of the network is uncertain. Furthermore, network performance in WSNs is a trade-off between longevity and performance.

A. Motivation

The researchers used the energy harvesting (EH) technique to avoid the problems caused by the sensor node's finite lifetime [8]. The EH process uses solar, radio frequency, wind, mechanical, and thermal energy resources to recharge the sensor nodes' batteries. This method provides an infinite energy supply to the sensor node at a significantly lower cost. Furthermore, solar harvesting devices work with various practical components, including solar panels, dc–dc regulators, energy predictors, maximum power point tracking (MPPT), and storage classes. The ability to store energy from external sources distinguishes solar-based harvesting nodes. Each sensor node is primarily a type of storage. Energy can be stored in networks using two types of devices that use a chemical process: rechargeable batteries and super-capacitors. The rate at which the sensor nodes can replenish their energy stores, on the other hand, is determined by the region in which they are located [9]. This applies to solar elevation angle, sunshine duration, and other weather-related conditions. In a WSN, sensing occurs in real-time, with continuous data gathering and processed data forwarding. Cycling control packets and sending duplicate data packets throughout the network produces congestion and depletes network resources. Excessive sensing within the network also results in frequent communication and the consumption of a substantial percentage of sensing cycles. As a result, sensor energy quickly depletes, and the network reshapes as clusters change rapidly [10]. These methods significantly impact resource availability and waste important sensing cycles on redundant and meaningless data packets. As a result, a system for effective synchronization between the CH and sensor nodes and scheduling.

B. Contribution

Considering the challenges in mind, in this work, we aim to design an efficient routing awareness-based scheduling (ERAS) strategy for controlling unnecessary data transmission in sensor networks. The ERAS approach mainly strives to enhance the battery-life capacity and network lifetime of the WSN. The primary objective of this study can be summed up as follows, according to our recapitulation.

1) We present a new clustering-based routing protocol known as ERAS, which uses a solar harvesting system in EH-WSNs.
2) The ERAS has been classified in various ways, including the initial phase, the setup phase, the scheduling phase, and the data transmission phase.
3) The CH is chosen based on the sum of the current round's leading residual energy and harvesting rate.
4) Following the selection of the CH, synchronization-based scheduling is implemented as TDMA in order to conserve intra-energy between the CH and the cluster members (CMs).
5) Finally, data is sent to the base station (BS) employing single-hop or multihop routing, and the suggested technique provides appropriate performance in terms of a perpetual network for EH-WSNs.

The remaining part of this article is structured as follows. In Section II, we have studied a related work based on cluster-based routing techniques, and preliminaries are followed in Section III. Section IV introduces and proposes an algorithm with various support functions. Moreover, it experimented with result performance in Section V, and the work conclusion showed in Section VI.

II. Related Work

This section studied various clustering-based routing protocols in sensor networks. Peng et al. [11] have presented a neutral clustering protocol in EH-WSNs named the neutral energy clustering (ENC) approach. The ENC method works distributively and elects one CH group (CHG). Zhang et al. [12] have proposed an extension of ENC to

Authorized licensed use limited to: Indian Institute of Information technology Sricity. Downloaded on January 19,2026 at 12:53:14 UTC from IEEE Xplore. Restrictions apply.

group the network. The authors introduced extending network lifetime by submitting the EH system. There is a suggestion for a way to make the network last longer by making EH nodes act as dedicated relay nodes for CHs. This method works in two ways: single cluster and multiple cluster formation in EH-WSNs. The work performance is better in various fields, just as it maximizes network lifetime, first node dead (FND), half node dead (HND), and last node dead (LND). Bozorgi et al. [13] have designed a cluster-based routing method in solar harvesting systems, and it is called novel energy-efficient clustering (NEEC). This method uses a new hybrid way to combine both static and dynamic clustering protocols.

Ahmed et al. [14] have presented an efficient clustering technique for the robust sensor networks called sleep-wake energy-efficient distributed (SEED). The CHs of the elevated energy region are articulating the BS through a longer distance. The SEED protocol has been categorized in various ways: initialization phase, CH selection, and data transmission phase. Amgoth and Jana [15] have developed an efficient routing technique known as energy routing awareness (ERA) in WSNs. This method is not as effective as other ways in terms of the lifetime of the network, the power imbalance factor, and the data aggregation it provides. To address this issue, Muthukumaran et al. [16] have designed a low-energy clustering approach using multilayer routing in WSNs. This protocol uses different routing methods, namely energy-efficient clustering, energy-efficient clustering, and hierarchical routing multilevel. Li and Liu [17] have proposed a discrete particle swarm optimization (DPSO) technique to perform clustering. The BS collects status data from each node and applies a variant of the DPSO algorithm to determine the optimal configuration for the sensor network. They have designed a cluster-based routing technique that used gateway clustering energy-efficient centroid (GCEEC) [18]. The work performance is based on the agriculture sector to improve various metrics in WSNs. Heinzelman et al. [19] have demonstrated one of the oldest and most famous algorithms called low energy adaptive clustering hierarchy (LEACH) in WSNs. The cluster-based routing protocol is employed to make this strategy work, and the CH is picked based on the node with the most incredible energy left over in the current round. Liu et al. [20] have created a more energy-efficient version of the LEACH methodology to increase network energy efficiency. Kuila and Jana, [21] have performed energy-efficient clustering-based routing algorithms using linear/nonlinear programming (LP/NL). The routing technique uses a multiobjective fitness function and an efficient particle encoding strategy (PSO). Tao et al. [22] have designed a flow-balanced routing protocol for sensor networks. This strategy minimizes power while maintaining coverage. The authors have defined the routing awareness technique [23]. Mainly, this work is based on underwater sensor networks and the triangle metric-based multilayered routing protocol (TM2RP) method. In [24], they demonstrated the cross-layer optimization technique EH-WSNs. Moreover, it uses one famous approach called cross-layer design with energy scavenging and transfer capabilities (CREST), and this method works clustering-based approach.

III. Preliminaries

In this section, we discuss the functionalities of the proposed work. The proposed ERAS strategy is divided into various models and a problem statement as below. At first, we consider some fundamental premises of the ERAS algorithm in EH-WSNs.

1) A set of sensor nodes $S = \{S_1, S_2, \ldots, S_n\}$ has been randomly deployed in 2-D network of size $(A \times A)$.
2) The position of the sink node and sensor nodes is stationary after the deployment in the monitoring area.
3) Each sensor node has the same communication range R_{com}.
4) A set of sensor nodes $C = \{S_1, S_2, \ldots, S_k\}$ serving as CH indicated by $f_{\text{ch}}(S_i) = 1, \forall S_i \in C$
5) The sensor network is configured based on each node's initial energy allocation level E_0.
6) The sensor node can accommodate the communication power control according to the distance of the receiver sensor nodes.
7) CH makes their CMs' schedules utilizing the TDMA schedule method.
8) Each node has a solar harvesting kit with an effective rechargeable battery.
9) Each sensor node has a different harvesting rate, which ranges between $[E^h_{\min}, E^h_{\max}]$.
10) For various load operations, building a cluster, routing, and other network performance, the EH rate is higher than that of the energy-consuming rate.
11) The complete operation of the sensor network is separated into the numeral of rounds r.

A. Energy Model

We divide the EH model into two categories: radio energy and solar EH. Various network load operations can define the energy model, and the sensor node energy spend rate is determined by three primary states: sensing, receiving, and transforming data. Furthermore, other conditions, such as a change in node location, weather change, and several dissimilar situations, may have occurred. Nomenclature shows the notations of the article.

1) Radio Energy Model: It is one of the most fundamental experiment domains of the LEACH protocol in WSNs, introduced by Heinzelman et al. [19]. The model is divided into two parts: free-space and multipath. Energy is dissipated from the sensor node battery to execute transmitting or obtaining packets in sensor networks. If a sensor node's transmission distance is less than the threshold value d_0, the free space model $\mathcal{J}_{\text{fs}}$ (d^2) is used; otherwise, a multipath model $\mathcal{J}_{\text{mp}}$ (d^4) is used. The value of the distance parameter is 87 m same as the LEACH model and it is calculated from $d_0 = (\mathcal{J}_{\text{fs}}/\mathcal{J}_{\text{mp}})^{1/2}$. Furthermore, the ratio of the power amplifier energy dissipation rate to total energy consumption can be determined as follows:

$$\text{ratio}(E) = \begin{cases} \dfrac{l\mathcal{J}_{\text{fs}}d^2}{\text{lE}_{\text{elec}} + l\mathcal{J}_{\text{fs}}d^2}, & \text{if } d \le d_0 \\ \dfrac{l\mathcal{J}_{\text{mp}}d^4}{\text{lE}_{\text{elec}} + l\mathcal{J}_{\text{mp}}d^4}, & \text{if } d > d_0. \end{cases} \tag{1}$$

Authorized licensed use limited to: Indian Institute of Information technology Sricity. Downloaded on January 19,2026 at 12:53:14 UTC from IEEE Xplore. Restrictions apply.

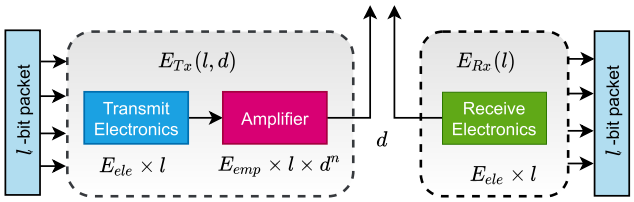


Fig. 1. Architecture of the radio energy dissipation model in WSNs.

The formulation of the radio energy model is shown as follows:

$$E_{T_x} = \begin{cases} \mathrm{lE_{elec}} + l\mathcal{J}_{\mathrm{fs}} \times d^2, & \text{if } d < d_o \\ \mathrm{lE_{elec}} + l\mathcal{J}_{\mathrm{mp}} \times d^4, & \text{if } d \geq d_o \end{cases} \tag{2}$$

where, l is the bit length of the packet and E_{elec} is the dissipated energy of the electronics circuit. The power amplifier is used to invert the loss of power control $d^n : n = 2$ to 4.

The radio energy is depleted to receive with l length of the bits packet, and it shows in the following equation:

$$E_{R_x} = l \times E_{\mathrm{elec}}. \tag{3}$$

Fig. 1 shows the basic architecture of the radio energy model.

2) Solar EH Model: The solar harvesting model is based on weather monitoring and it is classified into three types: sunny days, cloudy days, and mixed days. Today's sunny morning is immediately followed by cloudy noontime, which is referred to as a mixed day in EH-WSNs. The weather fluctuation affects the sensor node's EH rate. However, some energy from the solar panel is delivered to the sensor node in each round r [25]. The energy of ith node at the beginning of rth round $E(S_i, r)$ can be derived as follows:

$$E(S_i, r) = E_{\mathrm{res}}(S_i, r) + E_{\mathrm{har}}(S_i, r-1) \tag{4}$$

$$E_{\mathrm{har}}(S_i, r-1) = \eta_i \Delta_t \tag{5}$$

$$\eta_i = \mathrm{rand}\left(P_{h,\min}(r-1), P_{h,\max}(r-1)\right) \tag{6}$$

where, $E_{\mathrm{res}}(S_i, r)$ is the residual energy and $E_{\mathrm{har}}(S_i, r-1)$ justifies the harvesting rate within round r. Δ_t recognizes the time associated during the round and η_i is based on the EH-rate of ith node, which is a kind of random variable with uniform distribution lies between $P_{h,\min}(r-1)$ and $P_{h,\max}(r-1)$. Moreover, if the weather is sunny, the value of η_i is large, and for maximum and minimum value would be between $P_{h,\min}(r-1)$ and $P_{h,\max}(r-1)$.

B. Communication Model

In a network of N active sensor nodes, k CHs, and a BS, the communication between two nodes (S_i, S_j) refereed as $\mathrm{Comm}(S_i, S_j)$ follows the following communication policy.

1) If $(S_i, S_j) \in S$ then $\mathrm{Comm}(S_i, S_j)$ is valid iff $S_j \in C$.
2) If $(S_i, S_j) \in C$ then $\mathrm{Comm}(S_i, S_j)$ is valid iff $\mathrm{dist}(S_i, S_j) \leq R_{\mathrm{com}}$.
3) If $S_i \in S$ and $\mathrm{dist}(S_i, \mathrm{BS}) \leq R_{\mathrm{com}}$ then $\mathrm{Comm}(S_i, \mathrm{BS})$ is not allowed.
4) If $S_i \in C$ and $\mathrm{dist}(S_i, \mathrm{BS}) \leq R_{\mathrm{com}}$ then $\mathrm{Comm}(S_i, \mathrm{BS})$ is valid.

C. Problem Statement

Following the deployment of nodes across the monitoring area, long-term network sustainability is required, which is frequently measured by network lifetime. From the energy model, it can be concluded that communication consumes the most energy rather than sensing or processing. As a result, it is critical to reduce the number of communicating nodes while finding a path between each sensor node and the BS that considers both routing path quality and energy consumed by routing nodes. An EH module should be installed over sensor nodes to achieve a continuous network lifetime, where the energy consumed will be harvested using advanced harvesting methods. Moreover, the routing process can be better managed by following the clustering method. The problem of network lifetime maximization described above is separated into two subtasks: optimal CH search and optimal route discovery. As a result, the issue is divided into two phases. First, identify the best CH set CHbest to cover all sensor nodes. Then, to further enhance network lifetime, use the CHbest set to identify the most energy-efficient routing path for CHs. The following is a definition of the optimal node set search problem:

$$\min \sum_{i=1}^{|S|} f_{\mathrm{ch}}(S_i) \tag{7a}$$

$$\text{s.t.: } f_{\mathrm{ch}}(S_i) = \{0, 1\} \quad \forall i, \ 1 \leq i \leq |S| \tag{7b}$$

$$\sum_{i=1}^{|S|} \alpha_{j,i} \times f_{\mathrm{ch}}(S_i) \geq 1 \quad \forall j \in S, \ \alpha_{j,i} \in \{0, 1\} \tag{7c}$$

$$E_{\mathrm{consumed}}(S_i) \leq E_{\mathrm{har}}(S_i). \tag{7d}$$

When $f_{\mathrm{ch}}(S_i) = 1$, the sensor node S_i is CH, otherwise 0, if it is a normal node. $\alpha_{j,i} = 1$ indicates that S_j is present in the cluster range of CH S_i. The optimization problem solution is CHbest set with the objective $\min \sum_{i=1}^{|S|} f_{\mathrm{ch}}(S_i)$ following several constraints as (7b, 7c, 7d). The status of the node is presented by constraint (7b), while (7c) indicates that at least one CH is there covering the network. The constraint (7d) implies that the harvested energy in a round should be more than the energy consumed by an active sensor node S_i.

IV. Proposed ERAS Method

This section depicts the proposed ERAS algorithm, which is a cooperative clustering-based routing protocol. The algorithm executes in the following phases: initial, setup, scheduling, and data communication. The working steps of the ERAS method are explained below.

A. Initial Phase

This phase kept various types of information, namely the number of neighbor nodes, distance value, energy status, and radius calculation. Initially, the BS collects all essential data from sensor nodes and broadcasts an initial request as Reqst−Msg (ID, X_{BS}, Y_{BS}) to all nodes in the sensor network. Here the message window contains the personal node id number (ID) number and location of BS for all nodes in networks. Then each sensor node receives the BS request message, and immediately it replies with the message window as

Authorized licensed use limited to: Indian Institute of Information technology Sricity. Downloaded on January 19,2026 at 12:53:14 UTC from IEEE Xplore. Restrictions apply.

Self − Info − Msg(ID, location, E_{level}). The message contains its ID, location of the deployed node, and energy status. Moreover, after establishing a relationship between the sensor node and the BS, it is significant for further process, similar to the election of CH, data communication, etc. The distance parameter, on the other hand, is determined between the maximum and lowest distances from the BS and is dependent on the signal strength of the sensor nodes. The distance calculation is shown in the following equation:

$$d_s = d_{\max} - d_{\min} \tag{8}$$

where d_s is the distance between the BS and the farthest nodes. $d_{\max}$ is the farthest from BS, and $d_{\min}$ is the closest distance. Another function is related to the sensor node's energy level. It is essential for complete networks because all network load operations depend on the energy level. This effect is also related to the network lifetime. It has been calculated as energy level E_{level} of the sensor node S_i within round r

$$E_{\text{level}}(S_i, r) = E_{\text{initial}}(S_i, r-1) - E_{\text{diss}}(S_i, r-1) + E_{\text{har}}(S_i, r-1) \tag{9}$$

where, $E_{\text{initial}}(S_i, r-1)$ is initial energy of storage classes, $E_{\text{diss}}(S_i, r-1)$ depends on the energy consumption rate through various operations as $E_{\text{sensing}}/E_{\text{receiving}}/E_{\text{transfer}}$ of data and $E_{\text{har}}(S_i, r-1)$ identified how much energy harvested in different round r.

B. Setup Phase

This section describes the CH selection process, which is dependent on numerous parameters. In the ERAS algorithm, each sensor node S_i chooses expected number of CH based on the sum of its highest residual energy and EH rate for the current round. Then, the equation CH setup is as follows:

$$E[\#\text{CH}] = \sum_{S_i=1}^{N} \frac{E_{\text{res}}(S_i, r) + E_{\text{har}}(S_i, r-1)}{\sum_{i=1}^{N} E_{\text{res}}(S_i, r) + E_{\text{har}}(S_i, r-1)} \tag{10}$$

where N is the number of sensor nodes, $E_{\text{res}}(S_i, r)$ is the residual energy and $E_{\text{har}}(S_i, r-1)$ depends on the EH-rate of node ith with r rounds. Moreover, cluster formation is done through different control messages, especially advertisement (ADV) and join-request messages (Join-REQ). Once the sensor nodes have chosen CH using (10). The CH uses the nonpersistent carrier sense multiple access (CSMA) medium access control (MAC) protocol to send an ADV message to make an announcement. Each CM decides the CH with the least amount of communication energy to pick its cluster for the round. The strength of the signals received from each CH is used to make this conclusion. After that, each CM will once again make a request to join its CH via a nonpersistent CSMA protocol. Algorithm 1 shows the process of cluster formation.

C. Data Communication Phase

Data communication between sensor nodes is another critical step of the proposed ERAS strategy. This phase is divided into two parts: the scheduling and data transfer phases.

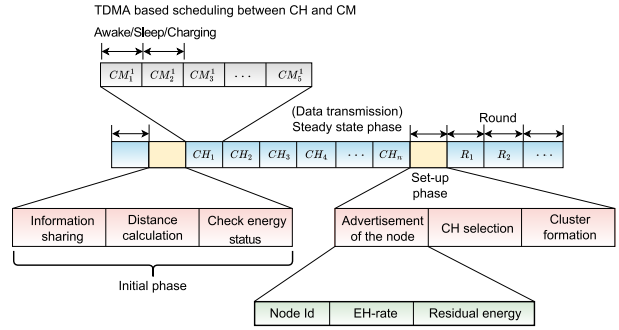


Fig. 2. Routing procedure of ERAS algorithm in EH-WSNs.

1) Scheduling Phase: After deciding on a CH node, it uses a TDMA-based schedule technique. Generally, the CH node uses the TDMA schedule for two reasons: 1) to prevent collision and 2) to preserve the intra-energy among the nodes. The first factor is based on a message transceiver between CH and CM and uses a fixed scheduling method to avoid data collision. The second factor depends on the sleep/wake scheduling technique. The radio components of each CM is turned off at all time except during their transceiver time. Then, it can be saved intra-energy resources of the sensor node. This process is also referred to duty cycle technique.

An Illustrative Example: We have taken a practical example of the TDMA-based scheduling method. Fig. 2 shows the complete scheduling approach between CH and CMs, and the execution process is made based on the synchronization manner and means (fixed time slot for each CM) in the cluster-based routing method. Here, $\{\text{CH}_1, \text{CH}_2, \ldots, \text{CH}_k\}$ denotes a set of CHs and CM_j^i denotes the jth CM under the CH CH_i. Further, each CM takes 5 s for an active node to complete their work, and the rest of the members would be in sleep mode. After completing the first CM, the second CM would come to schedule. This process will go on until the complete execution of all the CM. While CM is in sleep mode at that time, all members recharge using the solar harvesting technique. Here, CH schedules with five CMs based on fixed time slots.

1) The active time of the CM_1 = {(CM_1, active, 5 s) (CM_2, sleep, 5 s) (CM_3, sleep, 5 s) (CM_4, sleep, 5 s) (CM_5, sleep, 5 s)}.
2) The active time of the CM_2 = {(CM_1, sleep, 5 s) (CM_2, active, 5 s) (CM_3, sleep, 5 s) (CM_4, sleep, 5 s) (CM_5, sleep, 5 s)}.
3) The active time of the CM_3 = {(CM_1, sleep, 5 s) (CM_2, sleep, 5 s) (CM_3, active, 5 s) (CM_4, sleep, 5 s) (CM_5, sleep, 5 s) }.
4) The active time of the CM_4 = {(CM_1, sleep, 5 s) (CM_2, sleep, 5 s) (CM_3, sleep, 5 s) (CM_4, active, 5 s) (CM_5, sleep, 5 s) }.
5) The active time of the CM_5 = {(CM_1, sleep, 5 s) (CM_2, sleep, 5 s) (CM_3, sleep, 5 s) (CM_4, sleep, 5 s) (CM_5, active, 5 s)}.

The above process shows TDMA-based scheduling. Thereby, sensor nodes would save their extra energy resources. It is a highly effective method in the routing-based routing approach. Further, the node's residual energy is calculated

Authorized licensed use limited to: Indian Institute of Information technology Sricity. Downloaded on January 19,2026 at 12:53:14 UTC from IEEE Xplore. Restrictions apply.

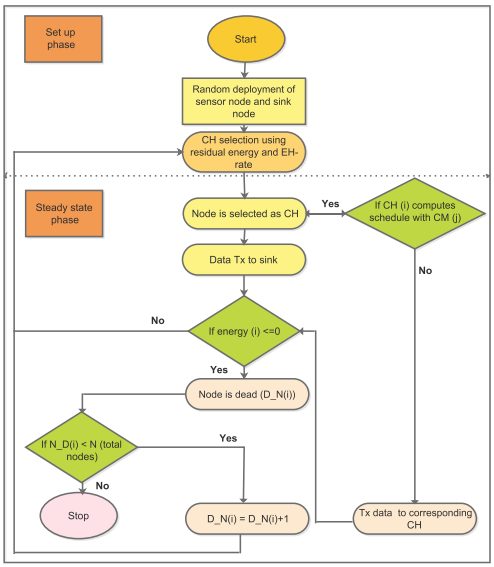


Fig. 3. Complete flowchart of ERAS algorithm in EH-WSNs.

using the standard sleep/awake method as follows:

$$E_{\text{res}}(S_i, r) = E_{\text{initial}}(S_i, r-1) - E_{\text{active}}(S_i, r-1) - E_{\text{sleep}}(S_i, r-1) + E_{\text{har}}(S_i, r-1) \quad (11)$$

where $E_{\text{active}}(S_i, r-1)$ shows the energy consumption during active operation and $E_{\text{sleep}}(S_i, r-1)$ denotes the sleep operation (all most few joule/sec). The energy depleted by a CH from any parts for the TDMA slot allocation is shown as

$$E_{\text{CH}}^{\text{TDMA}} = 2 \times l \times E_{\text{elec}} + l\mathcal{J}_{\text{fs}} \times d_{\text{toCM}}^2. \quad (12)$$

Now, the energy spent by a CM to obtain a TDMA slot is calculated as follows:

$$E_{\text{CM}}^{\text{TDMA}} = \text{lE}_{\text{elec}}. \quad (13)$$

Thus total energy consumption for TDMA schedule allocation is as follows:

$$E_{\text{total}}^{\text{TDMA}} = k \left\{ \underbrace{E_{\text{CH}}^{\text{TDMA}}}_{\text{Cluster Head}} + \underbrace{(N/k - 1)\, E_{\text{CM}}^{\text{TDMA}}}_{\text{Non Cluster}} \right\} \quad (14)$$

where, N is number of sensor nodes and k identifies cluster numbers, respectively.

2) Data Transfer: After the scheduling phase, CH uses signal processing to combine the data into a single signal. This process is called the data aggregation method to reduce unnecessary details or duplicate data. However, CH collects data from the entire CMs, and data is sent to the BS using the routing technique, as illustrated in Fig. 3. This phase is separated into sub-phases related to the data routing, namely intra- and inter-clustering-based routing. It is shown in (15) and (16) that data is routed from CH to BS and from CM to CH. This scattering occurs because each CH takes the CM signal, adds information about how it was combined, and then transmits it to the BS. Due to the distance between the BS and sensor nodes, multipath mode (d^4 power loss) is used

$$E_{\text{CH}} = \text{lE}_{\text{elec}}\left(\frac{N}{k} - 1\right) + \text{lE}_{\text{DA}}\left(\frac{N}{k}\right) + \text{lE}_{\text{elec}} + l\mathcal{J}_{\text{mp}} \times d_{\text{toBS}}^4 \quad (15)$$

where l denotes the total number of bits contained in the data packets, E_{DA} is an abbreviation for the data accumulation method, and k stands for the total number of clusters. The d_{toBS} distance between CH and BS is often considerable. Each CM delivers data to the CH, and the distance between CM and CH is often extremely short, known as free space distance (d^2 power loss)

$$E_{\text{CM}} = \underbrace{\text{lE}_{\text{elec}}}_{\text{Electronics Circuit}} + \underbrace{l\mathcal{J}_{\text{fs}} \times d_{\text{toCH}}^2}_{\text{Free Space}} \quad (16)$$

where d_{toCH} is the distance from CM to the CH. The energy consumption rate of the ERAS method depends on two operations especially, intra data routing and the inter-data routing technique. The intra-data routing phase is made up of other three associates given as follows:

$$E_{\text{Intra}}^{\text{routing}} = \underbrace{E_{\text{CMtoCH}}}_{\text{Transmission}} + \underbrace{E_{\text{CH}}^{\text{Reception}}}_{\text{Receipt}} + \underbrace{E_{\text{DA}}}_{\text{Aggregation}} \quad (17)$$

$$E_{\text{CMtoCH}} = \sum_{i=1}^{k} \sum_{j=1}^{m_i} E_{\text{Tx}}\left(\text{CM}_j, \text{CH}_i\right). \quad (18)$$

The energy consumption rate between CM and respective CHs is determined by (18). Where k is the number of clusters and m_i is the number of CM in the ith cluster. The $E_{\text{Tx}}(\text{CM}_j, \text{CH}_i)$ value reflects energy transfer from CM_j to its CH

$$E_{\text{CH}}^{\text{Reception}} = \sum_{i=1}^{k} (m_i \times E_{\text{Rx}}). \quad (19)$$

Equation (19) shows dissipated energy at the CH for receiving packets from its CMs here, m_i is the number of CMs associated with ith cluster and k is the number of CHs in the network

$$E_{\text{DA}} = \sum_{i=1}^{k} (l \times m_i \times E_{\text{DataBit}}) \quad (20)$$

where E_{DataBit} is the energy consumption rate for single-bit data aggregation and l represents the number of bits. Next, (21). is being used to determine inter-cluster data routing. The ERAS method's requirement is explained in Fig. 4(b) by a single-hop data transformation with no intermediate nodes. As a result, the rate of energy loss between the CH and the BS would be determined as follows:

$$E_{\text{Inter}}^{\text{routing}} = E_{\text{Tx}}\left(\text{CH}_i, \text{BS}\right). \quad (21)$$

Each phase process execution of the ERAS strategy is shown in Fig. 3. Similarly, the detailed steps of data routing are also presented in Algorithm 2 identifies the steady phase.

Lemma 1: The time complexity of the ERAS algorithm is $\mathcal{O}(N^2)$, where N is the number of nodes in the sensor network.

Authorized licensed use limited to: Indian Institute of Information technology Sricity. Downloaded on January 19,2026 at 12:53:14 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 ERAS: Set-Up Phase (Cluster Formation)

INPUT: Set of sensor node and sink node
OUTPUT: The list of cluster heads $CH_1, CH_2, CH_3, \ldots, CH_k$
Initialization of sensor nodes deployment process
The election process starts to choose CH among nodes
if $f_{ch}(S_i) = CH$ **then**
    Announced S_i is CH
end
for S_i **do**
    Broadcast control message with neighbor S_j
    Message header contains E_{res} and E_{har}
end
Select the CH election process for each round r
for <*each* CH_i> **do**
    <Calculate number of CHs through Eq. (10) for E_{res} and E_{har}>
    if *distance* $(d_i < d_{\mathcal{J}})$ **then** // It is based on minimum communication range to select CH
        CH_i broadcast ADV message using nonpersistent CSMA MAC protocol
    end
end
The coordination process executes between CH and CMs to elect number of CHs in various R_{com}
if CM_j *receives ADV message* **then**
    CM_j sends $JOIND$ message to the CH_i
end

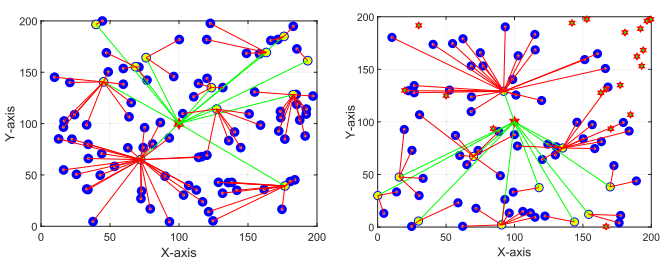


Fig. 4. Node deployment characteristics. (a) Initial round node deployment. (b) Deployment stage after several rounds with dead node.

Proof: The ERAS strategy contains several steps for determining the total complexity of the network. First, it estimates the distance of each node to another sensor node with the help of Euclidean distance. This process takes $\mathcal{O}(N^2)$ time. The number of rounds is another essential part of the ERAS method. Round is decided by the current energy state, defined as the initial energy of the sensor node battery, indicated by E_0, and the EH rate. For each round, the complexity of the entire method is $\mathcal{O}(N_r)$, where N_r is the number of rounds to select the number of CHs among N nodes. Each round produces approximate $\mathcal{O}(\log N)$ number CH, which takes $\mathcal{O}(N_r \log N)$ time. Therefore, the overall time complexity of the ERAS strategy becomes $\mathcal{O}(N_r \log N) + \mathcal{O}(N^2) = \mathcal{O}(N^2)$, as $N^2 \gg N_r \log N$.

Algorithm 2 ERAS: Steady Phase (Data Routing)

INPUT: List of CHs and CMs
OUTPUT: Data routing path via CHs
The identify number of cluster CH_k for particular rounds
for CH_k **do**
    if $d_k < d_{\mathcal{J}}$ **then**
        The elected number of CH_k is based on the Eq. (10) as sub-associative CH
    end
end
Scheduling process starts between CH and CM using TDMA
for CH_k **do**
    The allocate fixed time slots CM_j based on the TDMA schedule // Time for each CM to schedule 5*sec*
    CM_j transmits data to the CH_k in the assigned slots
end
The CH begins to broadcast the routing message in networks
if *Node* S_i *is the CH within its range* **then**
    CH_i induces TDMA schedule for proper message synchronization $slot_i$
end
$CH_i \leftarrow CM_j$ // The CH Gathers data packets from CM for data aggregation process
The data packet is sent to the BS using routing techniques
if *distance* $< d_o$ **then**
    Broadcast packet from CM_j to CH_i using Eq. (16) // Intra-clustering-based routing
else
    Broadcast packet from CH_i to BS using Eq. (15) // Inter-clustering-based routing using $CSMA$
end

Lemma 2: The message exchange complexity of the ERAS approach is $\mathcal{O}(1)$ for each individual node and $\mathcal{O}(N)$ for the total network with N number of nodes.

Proof: Using the CH ADV message and the JOIND message, each sensor node distributes control messages to build CH as part of the clustering-based routing protocol development process. As a result, it takes $\mathcal{O}(1)$ times for CH and CM to send and receive a message. Only a certain amount of time can be used to do this. On the other hand, the complete process takes $N - 1$ times as long as the number of sensor nodes involved in the cluster. Thus, the entire ERAS operation for cluster building takes at most $\mathcal{O}(N)$, which is the worst case scenario for message exchange.

Lemma 3: The time complexity of the scheduling-based routing algorithm is $\mathcal{O}(N)$ for sensor networks with N nodes.

Proof: In the scheduling-based routing phase, CH sets up a TDMA scheduling for their CMs for proper synchronization, and the complete execution process takes $\mathcal{O}(1)$. Each CH

Authorized licensed use limited to: Indian Institute of Information technology Sricity. Downloaded on January 19,2026 at 12:53:14 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PARAMETER USED FOR ERAS ANALYSIS

Parameter name	Value
Sensor network deployment	Random
BS location	(100,100)
E_{elec}	50 nJ/bit
E_{fs}	10 $pJ/bit/m^2$
E_{mp}	0.0013 $pJ/bit/m^4$
E_{DA}	5 $nJ/bit/message$
Data packet size	4000 bits
Number of sensor nodes	100 to 200
The initial energy rate E_0	0.5 J
The minimum value of EH-rate	0.025 J
The maximum value of EH-rate	0.23 J

TABLE II
SENSOR NETWORK SIZES

Sensor networks	# Sensor nodes	Networks size
$EH-WSN\#1$	100	200×200 m
$EH-WSN\#2$	200	500×500 m

must determine each sensor node's residual energy and verify the residual energy of $N-1$ sensors in the worst scenario. Therefore, the total time complexity to direct the data routing phase is $\mathcal{O}(N)$.

Remark #1: The ERAS algorithm's energy balance phenomena deliver the most efficient lifetime performance among the existing methods. Here, each round gets some extra energy from the ambient to the nodes. Thereby, it would be helpful for perpetual network operation. Generally, the energy dissipation rate is too high during the routing phase because some issues vary related to these factors, especially the distance between nodes, present-day weather conditions, and energy level E_0 of nodes. Therefore, the ERAS algorithm justifies a long-lasting and robust network lifetime in EH-WSNs.

V. PERFORMANCE EVALUATION

This section shows the performance analysis of our proposed ERAS technique, which is evaluated through the Matlab R2019b simulator tool. Here, we assumed the sensor network sizes are 200 × 200 and 500 × 500 m^2. For the accurate evaluation of this experiment, 100 to 200 sensor nodes are deployed randomly with some CHs (10%). We consider the energy dissipated from the electronic circuit to be 50 nJ/bit and for data aggregating 5 nJ/bit/message in WSNs. Moreover, the essential energy distribution E_0 is considered 0.5 J. The primary goal of the EH node is to maintain the charging level in various weather conditions and its operating duty cycle mode. In this way, the EH rate of the sensor node varies between 0.025 and 0.23 J. Tables I and II show the other related simulation parameters with various values. Further, to present the efficiency of the ERAS method, we have compared our technique with three standard algorithms, including hierarchical routing using multilevel (HRML) [16], ERA [15], SEED [14], TM2RP [23] and CREST [24].

A. Node Deployment Characteristics

The node deployment issue of sensor networks is critical because network performance depends on several factors.

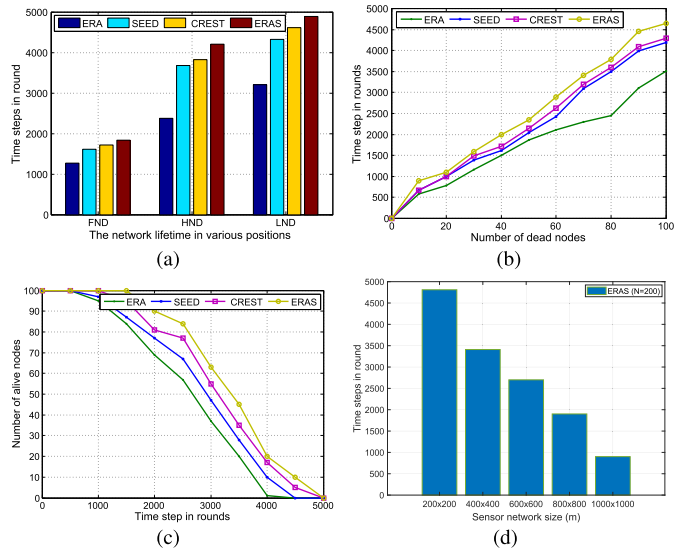


Fig. 5. Network lifetime analysis. (a) Network lifetime at various positions as FND, HND, and LND within rounds. (b) Network lifetime of dead nodes with standard techniques. (c) Number of alive nodes in standard algorithms. (d) Performance of ERAS strategy.

These factors can be divided into two ways, namely, random and deterministic. However, the proposed ERAS approach uses a random form for network deployment. It is also crucial because solar harvesting systems are associated with various weather monitoring conditions. Eventually, it affects the harvesting rate, such as the location of the deployed node, weather conditions, energy emission rate, and the angle of sunlight. These types of situations also affect perpetual network operation in EH-WSNs. The node deployment characteristics can be observed in Fig. 4(a) and (b). Where, Fig. 4(a) shows the initial round condition, and Fig. 4(b) identifies spending more rounds with the dead node.

B. Network Lifetime Performance

In sensor networks, the performance of the network lifetime is an essential component because it justifies the network's existence throughout time. This statistic has two parts: network lifetime and stability period. The network lifetime is defined as the period elapsed between the start of network operation till FND, LND, and HND of the currently functioning sensor nodes. On the other hand, the stability period is the duration from the beginning of network operation till the demise of the first sensor node. The perpetual network lifetime depends on the sensor node batteries and employs solar harvesting to receive extra energy from the environment. Fig. 5(a) justifies network lifetime ($\mathbb{T}_{\text{lifetime}}$) at various stages, such as FND, LND, and HND. Moreover, Fig. 5(b) shows the dead node condition, and Fig. 5(c) illustrates the sensor network stability ($\mathbb{T}_{\text{stability}}$) with alive nodes in every round. Here, we can notice in Fig. 5(d) that when the network size increases, the performance of network lifetime degrades at the same number of nodes ($N = 200$). Since this problem relies on data transfer through multihop routing, a greater amount of energy is required.

Authorized licensed use limited to: Indian Institute of Information technology Sricity. Downloaded on January 19,2026 at 12:53:14 UTC from IEEE Xplore. Restrictions apply.

TABLE III
LIFETIME OF SENSOR NETWORK AT VARIOUS ROUNDS IN EH-WSN#1

$EH-WSNs\#1$	ERA [15]	SEED [14]	CREST [24]	ERAS
FND	1270	1612	1712	1866
HND	2630	3611	3824	4223
LND	3210	4311	4612	4898
Average	2370	3178	3382.6	3662.3

TABLE IV
LIFETIME OF SENSOR NETWORK AT VARIOUS ROUNDS IN EH-WSN#2

$EH-WSNs\#2$	ERA [15]	SEED [14]	CREST [24]	ERAS
FND	1089	1232	1431	1766
HND	2134	2456	2567	2987
LND	2897	3211	3581	4011
Average	2040	2299.7	2666.7	2921.3

TABLE V
TIME COMPUTATION (ROUND) OF THE NETWORK [1000 × 1000] ($N = 200$)

Time (Round)	ERA [15]	SEED [14]	CREST [24]	ERAS
FND (r=300)	157	210	307	344
HND (r=600)	298	323	413	578
LND (r=900)	396	501	701	894

Now we define the network lifetime and network stability as follows:

$$\mathbb{T}_{\text{lifetime}} = \mathcal{T}_{\text{LND}} - \mathcal{T}_{\text{start}} \tag{22}$$

where, $\mathcal{T}_{\text{LND}} = r \times \text{LND}$, identifies the number of rounds still the death of all sensor nodes. Let $\mathcal{T}_{\text{start}}$ be the beginning time of network operation. Then we can define the network stability ($\mathbb{T}_{\text{stability}}$) as follows:

$$\mathbb{T}_{\text{stability}} = \mathcal{T}_{\text{FND}} - \mathcal{T}_{\text{start}} \tag{23}$$

where $\mathcal{T}_{\text{FND}} = r \times \text{FND}$, defines the number of rounds till the death of the first sensor node. The experimental analysis shows that the proposed ERAS method performs better than standard ERA, SEED, and CREST techniques. Figures also validate the network lifetime performance of ERA, SEED, CREST, and ERAS 67.3%, 73.8%, 75.2%, and 79.8%, respectively. Furthermore, Tables III and IV show the comparative analysis of network lifetime in EH-WSN#1 and EH-WSN#2 scenarios with standard algorithms. Moreover, Table V defines time computation (round) on the network lifetime performance in large network sizes at the same number of nodes ($N = 200$). A detailed analysis of alive sensor nodes (% wise) on various levels of EH-WSN is also presented in Table VI.

C. Packet Delivery Ratio

The packet delivery ratio (PDR) establishes how many packets have been sent to the BS. This process is associated with two operations. The first operation depends on how much data CH receives from various CM. The second part is based on received data transferred at the BS through the CH. This process is also referred to as the throughput of the network. The PDR performance also depends on two factors, especially the number of rounds and data aggregation. However, the ERAS technique employed a data aggregation approach with the help of (20) in EH-WSNs. The PDR for a sensor network can be calculated as follows:

$$\text{PDR} = \frac{\text{Packets sent from CM}}{\text{Packets received at the BS}}. \tag{24}$$

Regarding the above conditions, our proposed approach gives efficient performance compared to existing techniques, such as SEED, TM2RP, and CREST. Fig. 6(a) and (b) determine the number of data packets processed at the BS while preserving the EH rate. It comes from the fact that the packet delivery rate increases with our ERAS technique.

TABLE VI
PERCENTAGE OF DEAD NODE IN EH-WSN#1

Dead node (%)	ERA	SEED	CERST	ERAS
10	1723	1903	2101	2211
20	1921	2301	2298	2521
30	2210	2601	2710	2856
40	2456	2910	3211	3516
50	2710	3345	3511	3856
60	2891	3561	3723	4100
70	3017	3712	3902	4313
80	3110	3916	4213	4567
90	3216	4019	4413	4739
100	3312	4201	4610	4910

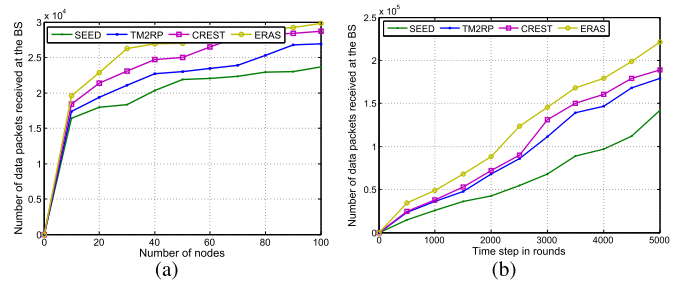


Fig. 6. PDR. (a) Number of data packets at the BS versus nodes. (b) Data packets at the BS within rounds in EH-WSN#2.

D. Sensor Node Performance

The performance of the sensor network can be determined using several networks and system-based parameters, namely average energy consumption rate in different rounds, residual energy, and energy levels in EH-WSNs. All the operations depend on the energy level of the sensor node battery for perpetual network performance. Here, the ERAS method calculates energy level as well as residual energy using (11). Fig. 7(a) justifies the performance of energy level at various energy values and Fig. 7(b) determined the average energy-consuming rate in different rounds. The average energy defines how much energy is consumed in total energy value. The average energy dissipated performance of HRML, SEED, and ERAS methods is 61%, 73%, and 81%, respectively, as presented in Fig. 7(b). It can be easily noted from the figures that the performance of the ERAS method is incremental.

E. Network Sustainability

Network sustainability identifies how much energy is harvested concerning energy consumption in the current round. Here, we have considered the energy consumption rate is less than or equal to the EH-rate in most of the round, and this relationship shows through (7d), which means condition

Authorized licensed use limited to: Indian Institute of Information technology Sricity. Downloaded on January 19,2026 at 12:53:14 UTC from IEEE Xplore. Restrictions apply.

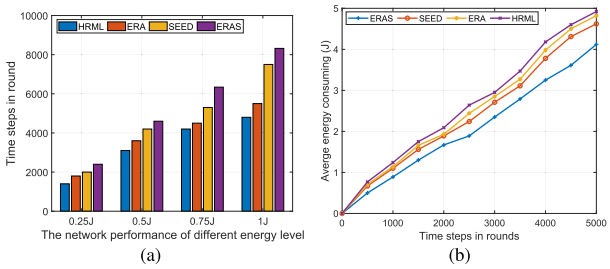


Fig. 7. Performance analysis. (a) Energy levels for sensor nodes. (b) Average energy consumption among the standard techniques.

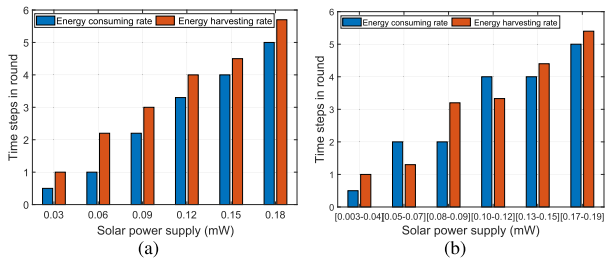


Fig. 8. Network sustainability. (a) EH rate in linear weather conditions. (b) EH rate in variable weather conditions.

$E_{\text{consuming}}(r) \leq E_{\text{har}}(r)$. However, it is a particular issue for long-term network sustainability. Sometimes, this justification is undecidable because the solar system is associated with multiple weather monitoring conditions. For example, if the weather is sunny, the EH rate is more than the energy-consuming rate. Otherwise, this process will be the opposite in most rounds r. According to the theory, we can observe both figures. Fig. 8(a) shows a sunny day EH rate with no environmental changes, which presents a linear harvesting process over time. Moreover, Fig. 8(b) justifies the cloudy weather forecasting system, which means the EH rate is lesser than the consuming rate in some rounds.

VI. Conclusion

Nowadays, the demand for sensory applications has increased in various places, such as weather monitoring, health monitoring, agriculture field, IoT devices, green computing, etc. But one general issue comes in a sensor network called the energy scarcity problem. It is remarkably often because all the network load operations depend on the battery's energy level. One of the best solutions is to overcome energy limitations using an EH system. Another method may be possible through clustering protocol in WSNs. We have designed a novel cluster-based routing ERAS technique related to both issues. The ERAS method is categorized into phases, incredibly the initial set-up, scheduling, and data routing phases. This algorithm uses a TDMA-based scheduling technique to avoid the collision of messages between CH and CM. We will extend our work with industrial sensor networks and incorporate deep reinforcement learning to optimize routing costs in the future.

References

[1] P. K. Donta, B. S. P. Rao, T. Amgoth, C. S. R. Annavarapu, and S. Swain, "Data collection and path determination strategies for mobile sink in 3D WSNs," *IEEE Sensors J.*, vol. 20, no. 4, pp. 2224–2233, Feb. 2020.

[2] M. Zheng, C. Wang, M. Song, W. Liang, and H. Yu, "SACR: A stability-aware cluster-based routing protocol for cognitive radio sensor networks," *IEEE Sensors J.*, vol. 21, no. 15, pp. 17350–17359, Aug. 2021.

[3] D. Sikeridis, E. E. Tsiropoulou, M. Devetsikiotis, and S. Papavassiliou, "Socio-physical energy-efficient operation in the internet of multipurpose things," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2018, pp. 1–7.

[4] A. Hazra, M. Adhikari, T. Amgoth, and S. N. Srirama, "A comprehensive survey on interoperability for IIoT: Taxonomy, standards, and future directions," *ACM Comput. Surveys*, vol. 55, no. 1, pp. 1–35, Jan. 2023.

[5] N. Mazumdar, A. Nag, and S. Nandi, "HDDS: Hierarchical data dissemination strategy for energy optimization in dynamic wireless sensor network under harsh environments," *Ad Hoc Netw.*, vol. 111, Feb. 2021, Art. no. 102348.

[6] M. S. Bahbahani and E. Alsusa, "A cooperative clustering protocol with duty cycling for energy harvesting enabled wireless sensor networks," *IEEE Trans. Wireless Commun.*, vol. 17, no. 1, pp. 101–111, Jan. 2018.

[7] A. Hazra, M. Adhikari, T. Amgoth, and S. N. Srirama, "Fog computing for energy-efficient data offloading of IoT applications in industrial sensor networks," *IEEE Sensors J.*, vol. 22, no. 9, pp. 8663–8671, May 2022.

[8] D. Ma, G. Lan, M. Hassan, W. Hu, and S. K. Das, "Sensing, computing, and communications for energy harvesting IoTs: A survey," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 2, pp. 1222–1250, 2nd Quart., 2020.

[9] D. K. Sah and T. Amgoth, "Renewable energy harvesting schemes in wireless sensor networks: A survey," *Inf. Fusion*, vol. 63, pp. 223–247, Nov. 2020.

[10] D. K. Sah, A. Hazra, R. Kumar, and T. Amgoth, "Harvested energy prediction technique for solar-powered wireless sensor networks," *IEEE Sensors J.*, vol. 23, no. 8, pp. 8932–8940, Apr. 2023.

[11] S. Peng, T. Wang, and C. P. Low, "Energy neutral clustering for energy harvesting wireless sensors networks," *Ad Hoc Netw.*, vol. 28, pp. 1–16, May 2015.

[12] P. Zhang, G. Xiao, and H.-P. Tan, "Clustering algorithms for maximizing the lifetime of wireless sensor networks with energy-harvesting sensors," *Comput. Netw.*, vol. 57, no. 14, pp. 2689–2704, Oct. 2013.

[13] S. M. Bozorgi, A. Shokouhi Rostami, A. A. R. Hosseinabadi, and V. E. Balas, "A new clustering protocol for energy harvesting-wireless sensor networks," *Comput. Electr. Eng.*, vol. 64, pp. 233–247, Nov. 2017.

[14] G. Ahmed, J. Zou, M. M. S. Fareed, and M. Zeeshan, "Sleep-awake energy efficient distributed clustering algorithm for wireless sensor networks," *Comput. Electr. Eng.*, vol. 56, pp. 385–398, Nov. 2016.

[15] A. Tarachand and P. K. Jana, "Energy-aware routing algorithm for wireless sensor networks," *Comput. Elect. Eng.*, vol. 41, pp. 357–367, Jan. 2015.

[16] K. Muthukumaran, K. Chitra, and C. Selvakumar, "An energy efficient clustering scheme using multilevel routing for wireless sensor network," *Comput. Electr. Eng.*, vol. 69, pp. 642–652, Jul. 2018.

[17] J. Li and D. Liu, "DPSO-based clustering routing algorithm for energy harvesting wireless sensor networks," in *Proc. Int. Conf. Wireless Commun. Signal Process. (WCSP)*, Oct. 2015, pp. 1–5.

[18] K. N. Qureshi, M. U. Bashir, J. Lloret, and A. Leon, "Optimized cluster-based dynamic energy-aware routing protocol for wireless sensor networks in agriculture precision," *J. Sensors*, vol. 2020, pp. 1–19, Jan. 2020.

[19] W. R. Heinzelman, A. Chandrakasan, and H. Balakrishnan, "Energy-efficient communication protocol for wireless microsensor networks," in *Proc. 33rd Annu. Hawaii Int. Conf. Syst. Sci.*, Jan. 2000, p. 10.

[20] Y. Liu, Q. Wu, T. Zhao, Y. Tie, F. Bai, and M. Jin, "An improved energy-efficient routing protocol for wireless sensor networks," *Sensors*, vol. 19, no. 20, p. 4579, Oct. 2019.

[21] P. Kuila and P. K. Jana, "Energy efficient clustering and routing algorithms for wireless sensor networks: Particle swarm optimization approach," *Eng. Appl. Artif. Intell.*, vol. 33, pp. 127–140, Aug. 2014.

[22] Y. Tao, Y. Zhang, and Y. Ji, "Flow-balanced routing for multi-hop clustered wireless sensor networks," *Ad Hoc Netw.*, vol. 11, no. 1, pp. 541–554, Jan. 2013.

[23] A. M. Khasawneh et al., "Green communication for underwater wireless sensor networks: Triangle metric based multi-layered routing protocol," *Sensors*, vol. 20, no. 24, p. 7278, Dec. 2020.

[24] M. M. Afsar and M. Younis, "A load-balanced cross-layer design for energy-harvesting sensor networks," *J. Netw. Comput. Appl.*, vol. 145, Nov. 2019, Art. no. 102390.

[25] A. Kansal, J. Hsu, S. Zahedi, and M. B. Srivastava, "Power management in energy harvesting sensor networks," *ACM Trans. Embedded Comput. Syst.*, vol. 6, no. 4, p. 32, Sep. 2007.

Authorized licensed use limited to: Indian Institute of Information technology Sricity. Downloaded on January 19,2026 at 12:53:14 UTC from IEEE Xplore. Restrictions apply.